

THE $RO(C_3)$ -GRADED COHOMOLOGY OF A COMPLETE FLAG VARIETY

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ABSTRACT. Let C_3 act on the complete flag variety of its complex regular representation, and let R be the $RO(C_3)$ -graded cohomology of a point with constant Mackey functor coefficients $\mathbb{F}_3$. We compute the genuine cohomology ring as $R[x_1, x_2, x_3]/(e_1, e_2 + a^2, e_3)$, where a is the Euler class of the two-dimensional real irreducible representation λ and each x_i has degree λ . A splitting into six suspended $H\mathbb{F}_3$ -modules gives a comparison isomorphism with Borel cohomology in pure multiples of λ . This determines normalized Schubert lifts and their multiplication, including Euler corrections to the ordinary divisor formulas. We describe the kernel and image of restriction to the six fixed flags. The kernel is a nonzero square-zero ideal at the fixed-orbit level, while the image requires divisibility by a^2 and a^3 in addition to the usual edge conditions. We also specify the graded Mackey functor structure.

1. INTRODUCTION

The cohomology of a complete flag variety admits presentations in terms of Chern classes and Schubert classes. For a finite group acting on a flag variety, genuine equivariant cohomology also contains coefficient classes in virtual representation degrees. These classes can be annihilated by passage to Borel cohomology and by restriction to fixed points. Thus a Borel presentation alone does not determine the genuine ring.

We consider the complete flag variety of the complex regular representation of C_3 . Set

$$G = C_3 = \langle g \rangle, \quad V = \mathbb{C}[C_3] = \mathbf{1} \oplus \chi \oplus \chi^2, \quad X = \text{Fl}(V),$$

where $\chi(g) = \exp(2\pi i/3)$. Let $\lambda = \chi_{\mathbb{R}}$, $H = H\mathbb{F}_3$, and

$$R^\alpha = \widetilde{H}_G^\alpha(S^0; \mathbb{F}_3) = \pi_{-\alpha}^G H, \quad R = R^\star.$$

Throughout, $H_G^\star(Y)$ means unreduced cohomology of an unbased space, or equivalently reduced cohomology of Y_+ . The superscript $\star$ ranges over $RO(G) = \mathbb{Z}\{1, \lambda\}$. Without an underline, cohomology denotes its value at G/G . Write $a = e(\lambda) \in R^\lambda$.

Theorem 1.1. *There are classes $x_i \in H_G^\lambda(X)$ for which*

$$(1.1) \quad H_G^\star(X) \cong \frac{R[x_1, x_2, x_3]}{(e_1(x), e_2(x) + a^2, e_3(x))}, \quad |x_i| = \lambda.$$

Here e_j denotes the j th elementary symmetric polynomial. As an R -module the ring is free on $1, A, B, C, D, F$, of respective degrees $0, \lambda, \lambda, 2\lambda, 2\lambda, 3\lambda$, where

$$(1.2) \quad \begin{aligned} A &= x_1, & B &= x_1 + x_2 + a, & C &= x_1 x_2, \\ D &= x_1(x_1 + a), & F &= x_1 x_2(x_1 + a). \end{aligned}$$

These are lifts of the ordinary Schubert classes indexed by $s_1, s_2, s_1 s_2, s_2 s_1, w_0$, respectively. Equivalently,

$$(1.3) \quad H_G^\star(X) \cong \frac{R[A, B]}{(A^3 - a^2 A, A^2 - AB + B^2 + a(A + B))}.$$

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The coefficient ring in this theorem is the entire representation-graded ring, including its negative cone. The generators are defined by a comparison isomorphism in degree λ ; no genuine projective bundle theorem or general complex orientation is assumed. The first Schubert products are

$$(1.4) \quad A^2 = D - aA, \quad B^2 = C - aB, \quad AB = C + D.$$

The terms involving a disappear after forgetting the action. The remaining products appear in Table 3.

There are six fixed flags, indexed by S_3 . Their restriction map

$$i^* : H_G^*(X) \longrightarrow R^6$$

has a nonzero kernel. More precisely, put $J_r = \text{Ann}_R(a^r)$. We prove

$$(1.5) \quad \ker i^* = J_1A \oplus J_1B \oplus J_2C \oplus J_2D \oplus J_3F.$$

This ideal has square zero at G/G . Restriction becomes an isomorphism after inverting a . Before localization, its image is described by five explicit congruences, two modulo a , two modulo a^2 , and one modulo a^3 . The tuple $(0, 0, 0, 0, 0, a)$ satisfies every root-curve edge condition but fails the last congruence. This is a failure of the ordinary edge-divisibility description for this particular restriction map, even in a degree where the map is injective.

The Schubert representation filtration gives an H -module splitting. This yields a comparison with Borel cohomology in degrees $n\lambda$. A basis argument then establishes the resulting presentation over the full coefficient ring R .

Relation to previous work. Lewis computed representation-graded cohomology of projective spaces with linear C_p -actions using Burnside coefficients [Lew88]. Shulman's account supplies a corrected proof of joint detection by underlying cohomology and fixed-point restriction in even underlying degrees [Shu10, Theorems 3.10–3.11]. That result uses Burnside coefficients over a ground ring without p -torsion and is not applied to constant $\mathbb{F}_3$ coefficients here. Basu and Ghosh compute the latter coefficient ring for C_p and prove a freeness theorem for sparse representation-cell complexes [BG21, §3.2 and Theorem 4.12]. Their calculation is the coefficient input to our proof. More recent projective-space calculations with constant coefficients are given by Basu, Dey, and Karmakar [BDK25]. Dugger and Hazel study the cohomology of a point for odd-order cyclic groups with constant integral coefficients [DH26].

Genuine flag-space calculations also occur in Kronholm's dissertation [Kro08, Proposition X.2]: the spaces there are finite-length real flags in a complete C_2 -universe, with constant $\mathbb{F}_2$ coefficients. The ordinary and torus-equivariant Schubert constructions used below are classical; we follow the conventions explained in [Bri05, §1.2] and [Kaj15, §2]. We have not found an explicit computation of this particular genuine C_3 -equivariant flag-manifold cohomology ring in the literature. Our contribution is the explicit ring together with its Schubert multiplication and the kernel and image of fixed-point restriction. The comparison proposition is recorded as a consequence of the stated H -module splitting and the known coefficient calculation.

Sections 2–4 establish the coefficient conventions and the splitting. Section 5 proves the comparison statement and computes the Borel ring. Sections 6 and 7 prove Theorem 1.1 and compute Schubert products. Section 8 treats fixed-point restriction, and Section 9 gives the Mackey structure and ordinary specializations.

2. CONVENTIONS AND THE COEFFICIENT RING

We use cohomological representation degrees. The positive coefficient classes have degrees

$$(2.1) \quad |a| = \lambda, \quad |u| = \lambda - 2, \quad |\kappa| = \lambda - 1.$$

Choose u using the complex orientation of χ on restriction to the trivial subgroup. The same letter will denote its image in Borel cohomology. The element u is not invertible in R .

Here is the coefficient calculation of [BG21, Definition 3.3 and Proposition 3.4, pp. 39–41], expressed in our grading. There is a decomposition

$$(2.2) \quad R = P \oplus N, \quad P = \mathbb{F}_3[a, u] \otimes \Lambda(\kappa), \quad N = \bigoplus_{j,k \geq 1} \mathbb{F}_3\{\nu_{j,k}, \mu_{j,k}\},$$

where

$$(2.3) \quad |\nu_{j,k}| = 1 + 2j - (j+k)\lambda, \quad |\mu_{j,k}| = 2j + (1-j-k)\lambda.$$

We normalize the symbols so that

$$(2.4) \quad \begin{aligned} a\nu_{j,k} &= \nu_{j,k-1}, & u\nu_{j,k} &= \nu_{j-1,k}, & a\mu_{j,k} &= \mu_{j,k-1}, & u\mu_{j,k} &= \mu_{j-1,k}, \\ \kappa\nu_{j,k} &= \mu_{j,k}, & \kappa\mu_{j,k} &= 0. \end{aligned}$$

A symbol with a zero index means zero. Also $N^2 = 0$. The latter relation follows directly from the module description: every element of N is infinitely divisible by a and killed by a power of a . If $a^r n = 0$ and $n' = a^r m$ with $m \in N$, then $nn' = 0$. The symbols in (2.3) correspond to the suspended fractional symbols in the cited homotopy calculation after reversing degrees. They are not fractions in R .

For $\alpha = m + n\lambda$, the parity is that of its underlying dimension $m + 2n$. Thus a and u are even, whereas κ and $\nu_{j,k}$ are odd. All polynomial generators of the flag ring will commute with the coefficients.

Lemma 2.1. *In the above convention,*

$$(2.5) \quad R^{1-r\lambda} = 0 \quad (r \geq 0), \quad R^{r\lambda} = \begin{cases} \mathbb{F}_3 a^r, & r \geq 0, \\ 0, & r < 0. \end{cases}$$

For $r \geq 1$,

$$(2.6) \quad J_r = \text{Ann}_R(a^r) = \text{span}_{\mathbb{F}_3}\{\nu_{j,k}, \mu_{j,k} : j \geq 1, 1 \leq k \leq r\}.$$

Proof. A monomial $a^i u^j \kappa^\epsilon$ in P has degree $(-2j - \epsilon) + (i + j + \epsilon)\lambda$, with $i, j \geq 0$ and $\epsilon \in \{0, 1\}$. Its coefficient of the trivial representation cannot be 1, and can be 0 only when $j = \epsilon = 0$. The corresponding coefficient in the degree of a negative-cone generator is at least 2. This proves (2.5). Multiplication by a is injective on P . On N , its powers shift the second index as in (2.4), giving (2.6). $\square$

Choose $s \in H^1(BG; \mathbb{F}_3)$ so that $t = \beta s$ is the first Chern class of the line bundle associated to χ . Put

$$R_B^* = \widetilde{H}_G^*(EG_+) = H^*(BG; \mathbb{F}_3)[u^{\pm 1}] = \mathbb{F}_3[t] \otimes \Lambda(s)[u^{\pm 1}], \quad |s| = 1, \quad |t| = 2.$$

The coefficient comparison, with this orientation, is

$$(2.7) \quad q_R : R \longrightarrow R_B, \quad a \longmapsto ut, \quad u \longmapsto u, \quad \kappa \longmapsto us, \quad N \longmapsto 0.$$

This is the calculation in [BG21, §3.2, pp. 40–41]. In particular,

$$(2.8) \quad R_B = R[u^{-1}], \quad \ker q_R = N, \quad R[a^{-1}] = \mathbb{F}_3[a^{\pm 1}, u] \otimes \Lambda(\kappa).$$

For example $0 \neq \mu_{1,1} \in R^{2-\lambda}$ is killed by both a and u . The formulas for restriction and transfer of coefficients will be established in Section 9.

3. EQUIVARIANT SCHUBERT CELLS

Fix a basis e_1, e_2, e_3 of V on which $g = \text{diag}(1, \zeta, \zeta^2)$, where $\zeta = \exp(2\pi i/3)$, and write $E_i = \mathbb{C}e_i$. We compose permutations as functions: $(vw)(i) = v(w(i))$. In one-line notation,

$$s_1 = 213, \quad s_2 = 132, \quad s_1 s_2 = 231, \quad s_2 s_1 = 312, \quad w_0 = 321.$$

All tables use the order 123, 213, 132, 231, 312, 321.

3.1. Fixed flags and tangent representations. The three eigenvalues of g are distinct. Its spectral projections are polynomials in g , so every invariant subspace of V is a sum of eigenlines. It follows that the fixed flags are exactly

$$(3.1) \quad p_w = (E_{w_1} \subset E_{w_1} \oplus E_{w_2} \subset V), \quad w \in S_3.$$

At such a flag the complex tangent representation is

$$(3.2) \quad T_{p_w} X \cong \bigoplus_{i < j} \text{Hom}(E_{w_i}, E_{w_j}) = \bigoplus_{i < j} \chi^{w_j - w_i}.$$

Every summand is nontrivial, and the underlying real representation is 3λ at each fixed point. The individual complex weights are recorded in Table 1.

The real representations underlying χ and χ^2 are isomorphic. Complex conjugation identifies them and reverses the complex orientation. Their Borel Euler classes are t and $-t$. In the common degree λ using the chosen u , the corresponding oriented factors are a and $-a$. We retain these signs throughout.

3.2. Linear coordinates and the filtration. Let $\mathcal{B}$ be the upper triangular subgroup of $GL(V)$. The Schubert cell $\mathcal{B}p_w$ has unique echelon coordinates

$$(3.3) \quad v_i = e_{w_i} + \sum_{\substack{j > i \\ w_j < w_i}} z_{ji} e_{w_j}, \quad F_r = \langle v_1, \dots, v_r \rangle.$$

To obtain these coordinates one normalizes each pivot and subtracts earlier pivot columns; the remaining entries are precisely those indexed by inversions. Applying g and renormalizing each pivot gives

$$g \cdot z_{ji} = \zeta^{w_j - w_i} z_{ji}.$$

Thus the cell is equivariantly isomorphic, as an affine space, to the linear representation

$$(3.4) \quad W_w = \bigoplus_{\substack{i < j \\ w_i > w_j}} \chi^{w_j - w_i}.$$

For $w = 231$ the nonconstant columns are $e_2 + z_{31}e_1$ and $e_3 + z_{32}e_1$, with weights χ^2 and χ . For $w = 321$ they are $e_3 + z_{21}e_2 + z_{31}e_1$ and $e_2 + z_{32}e_1$, with weights χ^2, χ, χ^2 .

The Bruhat closure relation gives a closed filtration $X_k = \bigcup_{\ell(w) \leq k} \mathcal{B}p_w$ for $0 \leq k \leq 3$; see [Bri05, Proposition 1.2.1]. Its successive quotients are the one-point compactifications of the cells of a fixed length:

$$(3.5) \quad X_k / X_{k-1} \cong \bigvee_{\ell(w)=k} S^{W_w} \cong \bigvee_{\ell(w)=k} S^{k\lambda}.$$

For $k = 0$ this means $(X_0)_+ \cong S^0$.

TABLE 1. Complex tangent weights and Schubert representations. The tangent weights are ordered by $(1, 2), (1, 3), (2, 3)$.

w	Tangent weights	W_w	$(W_w)_{\mathbb{R}}$
123	(χ, χ^2, χ)	0	0
213	(χ^2, χ, χ^2)	χ^2	λ
132	(χ^2, χ, χ^2)	χ^2	λ
231	(χ, χ^2, χ)	$\chi \oplus \chi^2$	2λ
312	(χ, χ^2, χ)	$\chi \oplus \chi^2$	2λ
321	(χ^2, χ, χ^2)	$\chi \oplus 2\chi^2$	3λ

An independent check of the representation indices comes from the following invariant Morse function. Represent a flag by its ordered orthogonal lines, choose unit vectors v_i on them, and put

$$(3.6) \quad h(F) = 10\langle D_0 v_1, v_1 \rangle + 3\langle D_0 v_2, v_2 \rangle, \quad D_0 = \text{diag}(0, 1, 4).$$

Its critical-point equation is $[D_0, U \text{diag}(10, 3, 0)U^*] = 0$, so the critical points are the six p_w . The negative Hessian directions are exactly the pairs $i < j$ with $w_i > w_j$, and hence form W_w . The critical values occur in nondecreasing length. The proof of the classical Morse attachment theorem [Mil63, Theorem 3.2] adapts equivariantly here, using a G -invariant metric and equivariant Morse coordinates, and gives the same sphere cofibers.

4. ADDITIVE BREDON COHOMOLOGY

Proposition 4.1. *There is an equivalence of genuine G -equivariant H -modules*

$$(4.1) \quad H \wedge X_+ \simeq H \vee (\Sigma^\lambda H)^{\vee 2} \vee (\Sigma^{2\lambda} H)^{\vee 2} \vee \Sigma^{3\lambda} H.$$

Consequently, as $RO(G)$ -graded Mackey functors,

$$(4.2) \quad \underline{H}_G^\alpha(X) \cong \bigoplus_{w \in S_3} \underline{R}^{\alpha - \ell(w)\lambda}.$$

Proof. The first stage consists of one fixed point. Suppose the preceding stage, after smashing with H , is a wedge of modules $\Sigma^{j\lambda} H$ with $j \leq k$. The cofiber of the next attachment is $\Sigma^{k\lambda} H$. Desuspending its connecting map gives components in

$$(4.3) \quad [\Sigma^{k\lambda-1} H, \Sigma^{j\lambda} H]_{H\text{-mod}}^G = R^{1-(k-j)\lambda} = 0.$$

The equality follows from the free rank-one H -module adjunction, with $R^\alpha = \pi_{-\alpha}^G H$. The vanishing is Lemma 2.1; it also applies when $j = k$. Thus the entire connecting map is null in the genuine H -module category. Its cofiber triangle splits. Induction gives (4.1).

Apply the internal H -module dual $F_H(-, H)$. The free-module adjunction identifies $F_H(H \wedge X_+, H)$ with $F(X_+, H)$, and the dual of the finite wedge is the wedge of the $\Sigma^{-\ell(w)\lambda} H$. Taking all equivariant homotopy Mackey functors proves (4.2). $\square$

This proof gives more than degeneration of a cellular spectral sequence. For the length filtration, if the second index records total cohomological degree, the relative groups are

$$E_1^{k,\alpha} = \bigoplus_{\ell(w)=k} R^{\alpha-k\lambda}, \quad d_r : E_r^{k,\alpha} \longrightarrow E_r^{k+r,\alpha+1}.$$

On a relative generator each potential differential has coefficient in $R^{1-r\lambda}$; R -linearity makes all differentials zero. Equation (4.3) also eliminates additive extensions by splitting the underlying triangles. No splitting of X_+ into sphere spectra, and no multiplicative splitting, is asserted.

As a check on the grading, (4.2) gives

$$\dim_{\mathbb{F}_3} H_G^{n\lambda}(X) = 1, 3, 5, 6, 6, \dots \quad \text{for } n = 0, 1, 2, 3, 4, \dots$$

The powers of a account for the difference from the ordinary Betti numbers 1, 2, 2, 1.

5. COMPARISON WITH BOREL COHOMOLOGY

Write $Y_B = EG \times_G Y$ and $H_B^*(Y) = H^*(Y_B; \mathbb{F}_3)$. We distinguish this integer-graded ring from

$$H_G^*(EG \times Y) = H_B^*(Y)[u^{\pm 1}].$$

The identification follows from the oriented representation-sphere suspensions on a free G -space [BG21, Proposition 3.2]. In particular its component of degree $n\lambda$ is $u^n H_B^{2n}(Y)$.

We record the following consequence of finite H -module duality and (2.7).

Proposition 5.1 (Comparison in pure representation degrees). *Let Y be a finite G -CW complex such that*

$$(5.1) \quad H \wedge Y_+ \simeq \bigvee_{j=1}^m \Sigma^{k_j \lambda} H, \quad k_j \geq 0.$$

For every integer n , the natural comparison is an isomorphism

$$(5.2) \quad q_Y : H_G^{n\lambda}(Y) \xrightarrow{\cong} u^n H_B^{2n}(Y).$$

These maps preserve products. Under hypothesis (5.1), the comparison on the entire $RO(G)$ -graded ring is localization at u , and

$$(5.3) \quad \ker q_Y = NH_G^*(Y).$$

Proof. Put $C_Y = F(Y_+, H)$. Dualizing (5.1) gives $C_Y \simeq \bigvee_j \Sigma^{-k_j \lambda} H$. Write $\eta_C : C \rightarrow F(EG_+, C)$ for precomposition with $EG_+ \rightarrow S^0$. Under $F(EG_+, C_Y) \simeq F((EG \times Y)_+, H)$, the map η_{C_Y} is the cochain pullback along $EG \times Y \rightarrow Y$. For any chosen splitting $\phi : C_Y \xrightarrow{\sim} C' = \bigvee_j \Sigma^{-k_j \lambda} H$, naturality gives

$$F(EG_+, \phi) \circ \eta_{C_Y} = \eta_{C'} \circ \phi.$$

Finite wedges are finite products, so $\eta_{C'}$ is the direct sum of the summand comparisons. Function spectra also commute with suspension by an invertible representation sphere. In cohomological degree $n\lambda$ the j th component is therefore

$$R^{(n-k_j)\lambda} \longrightarrow R_B^{(n-k_j)\lambda}.$$

For $r = n - k_j \geq 0$ both groups are one-dimensional, and the map sends a^r to $u^r t^r$. For $r < 0$ both groups vanish. This proves (5.2). The original comparison is a map of cohomology rings, so these isomorphisms preserve products.

In arbitrary degrees the same direct-sum calculation identifies the map with coefficient localization on each summand. Since the sum is finite, it induces an isomorphism after inverting u , with kernel the sum of the shifted copies of N . This sum is precisely $NH_G^*(Y)$. $\square$

In particular, (5.2) applies to X by Proposition 4.1. It will be used to lift relations only in pure multiples of λ . Equation (5.3) precludes an identification of the full genuine ring with the Borel ring by regrading.

5.1. The ordinary flag bundle. Let $\mathcal{E}_i = EG \times_G E_i$ over BG . Then X_B is the complete flag bundle of

$$\mathcal{V} = \mathcal{E}_1 \oplus \mathcal{E}_2 \oplus \mathcal{E}_3 = \mathbf{1} \oplus \mathcal{L}_\chi \oplus \mathcal{L}_{\chi^2}.$$

On this flag bundle let $\mathcal{F}_i$ be the tautological subbundles and $L_i = \mathcal{F}_i / \mathcal{F}_{i-1}$ the tautological line quotients. Put $z_i = c_1(L_i^\vee)$. The original bundle has roots $0, t, 2t = -t$, so

$$c(\mathcal{V}) = (1 + 0)(1 + t)(1 - t) = 1 - t^2.$$

Its dual has roots $0, -t, t$ and the same total Chern class. Hence

$$(5.4) \quad e_1(z) = 0, \quad e_2(z) = -t^2, \quad e_3(z) = 0.$$

Proposition 5.2. *There is an isomorphism of $H^*(BG; \mathbb{F}_3)$ -algebras*

$$(5.5) \quad H_B^*(X) = \frac{\mathbb{F}_3[t, z_1, z_2, z_3] \otimes \Lambda(s)}{(e_1(z), e_2(z) + t^2, e_3(z))}, \quad |z_i| = |t| = 2, \quad |s| = 1.$$

A module basis over $H^(BG; \mathbb{F}_3)$ is*

$$(5.6) \quad 1, z_1, z_2, z_1 z_2, z_1^2, z_1^2 z_2.$$

Proof. First choose L_1 by passing to $\mathbb{P}(\mathcal{V})$, then choose L_2 by projectivizing the rank-two quotient $\mathcal{V}/L_1$. The classes $1, z_1, z_1^2$ restrict to a basis on the first projective fiber, and $1, z_2$ do so on the second. Two applications of ordinary Leray–Hirsch [Hat02, Theorem 4D.1] give (5.6). These statements apply with $\mathbb{F}_3$ coefficients; the tautological classes are defined on the entire bundles, so the fiberwise-basis hypothesis holds.

The Chern identities (5.4) give a map from the displayed quotient to cohomology. Eliminating z_3 identifies that quotient with the successive monic extensions

$$(H^*(BG; \mathbb{F}_3)[z_1]/(z_1^3 - t^2 z_1)) [z_2]/(z_2^2 + z_1 z_2 + z_1^2 - t^2).$$

It has exactly the basis (5.6), which maps to the Leray–Hirsch basis. The map is an isomorphism. $\square$

Every basis element in (5.6) is even. It follows that the even part of (5.5) consists of the terms without s . This observation will enter the definition of the genuine generators.

6. THE GENUINE COHOMOLOGY RING

6.1. Generators and a basis argument. By Proposition 5.1 in degree λ , there is a unique class $x_i \in H_G^\lambda(X)$ satisfying

$$(6.1) \quad q_X(x_i) = u z_i = u c_1^B(L_i^\vee).$$

Naturality of comparison gives

$$(6.2) \quad \text{res}(x_i) = v y_i, \quad y_i = c_1(L_i^\vee) \in H^2(X; \mathbb{F}_3), \quad v = \text{res}(u).$$

Here v records degree $\lambda - 2$ at the trivial subgroup; after forgetting the representation grading it is the unit in degree zero.

We will use a basis observation to pass from these particular degrees to all genuine degrees.

Lemma 6.1. *Suppose $b_1, \dots, b_m$ is a homogeneous R -basis of $H_G^*(Y)$ supplied by a splitting as in (5.1), with $|b_j| = k_j \lambda$. Let $c_j \in H_G^{k_j \lambda}(Y)$ have restrictions which, after removing the factors v^{k_j} , form an ordinary cohomology basis of Y . Then the c_j form an R -basis of $H_G^*(Y)$ and a free $\underline{R}$ -basis of the graded Mackey functor $\underline{H}_G^*(Y)$.*

Proof. Order both families by nondecreasing k_j . In $c_j = \sum_i d_{ij} b_i$, the coefficient d_{ij} lies in $R^{(k_j - k_i)\lambda}$. By Lemma 2.1, it is zero if $k_i > k_j$ and is a scalar multiple of $a^{k_j - k_i}$ otherwise. The change-of-basis matrix is therefore block triangular; its diagonal blocks have entries in $\mathbb{F}_3$.

The underlying $H\mathbb{F}_3$ -module splitting gives an ordinary basis from the b_i . Restriction kills the positive powers of a , so the diagonal blocks are precisely the ordinary degree-wise change-of-basis matrices. They are invertible over $\mathbb{F}_3$ by hypothesis. A finite block triangular matrix with invertible diagonal is invertible over R : after factoring out its diagonal part, invert the remaining $1 + T$ by the finite series $1 - T + T^2 - \dots$, since T is strictly block triangular. These operations are coefficient-module maps on both orbits. Equivalently they give an invertible matrix of maps between the free suspended H -modules dual to the chosen basis. Thus they also give a basis at the Mackey functor level. $\square$

6.2. Proof of the presentations.

Proof of Theorem 1.1. Under comparison, the three expressions $e_1(x), e_2(x) + a^2, e_3(x)$ map to $u e_1(z), u^2(e_2(z) + t^2), u^3 e_3(z)$, respectively. They vanish by Proposition 5.2. They have degrees $\lambda, 2\lambda, 3\lambda$, where comparison is injective. The genuine relations therefore hold.

Let Q denote the quotient in (1.1). After setting $x_3 = -x_1 - x_2$, put

$$f = x_2^2 + x_1 x_2 + x_1^2 - a^2, \quad h = x_1^3 - a^2 x_1.$$

Then $e_2 + a^2 = -f$ and $e_3 = h - x_1 f$. Thus

$$(6.3) \quad Q \cong (R[x_1]/(h)) [x_2]/(f).$$

Division by a monic polynomial works over an arbitrary coefficient ring, including one with zero divisors. Successive division in (6.3) gives the unique module normal forms

$$(6.4) \quad 1, x_1, x_2, x_1 x_2, x_1^2, x_1^2 x_2.$$

Their images in actual cohomology restrict, with the appropriate powers of v , to the ordinary basis obtained from the two projective bundles over a point. Lemma 6.1, applied to Proposition 4.1, shows that these images are an R -basis of $H_G^*(X)$. The map $Q \rightarrow H_G^*(X)$ therefore maps a basis to a basis and is an isomorphism. This excludes additional relations in every representation degree.

Define A, B, C, D, F by (1.2). The inverse changes for the nontrivial monomials are

$$x_1 = A, \quad x_2 = B - A - a, \quad x_1 x_2 = C, \quad x_1^2 = D - aA, \quad x_1^2 x_2 = F - aC.$$

Thus $1, A, B, C, D, F$ is another R -basis. Their Schubert normalization is proved geometrically in the next section. Finally substitute

$$x_1 = A, \quad x_2 = B - A - a, \quad x_3 = a - B.$$

The polynomial h becomes $A^3 - a^2 A$, and in characteristic three f becomes $A^2 - AB + B^2 + a(A + B)$. This invertible change of variables proves (1.3). $\square$

All products of the generators were detected in pure multiples of λ , but the resulting presentation is over the whole ring R . In particular, multiplication by a negative-cone coefficient is part of (1.1), despite being invisible under q_X .

7. EQUIVARIANT SCHUBERT CALCULUS

7.1. Geometric normalization. Use opposite Schubert varieties relative to the upper triangular cells of Section 3. Their complex codimensions are $\ell(w)$. Denote their Borel classes by S_w . The underlying ordinary classes form the usual Schubert basis [Bri05, §1.3]; the analogous torus-equivariant basis is described in [Kaj15, Theorem 2.2]. In this rank the required Borel representatives can be obtained directly from projections of the tautological bundles.

For s_1 the condition is $F_1 \subset E_2 \oplus E_3$. On the Borel flag bundle this is the zero locus of $L_1 \rightarrow \mathcal{E}_1$, so its divisor class is z_1 . For s_2 the condition is $E_3 \subset F_2$. It is the zero locus of the determinant of $\mathcal{F}_2 \rightarrow \mathcal{E}_1 \oplus \mathcal{E}_2$, giving $z_1 + z_2 + t$. These are regular divisor conditions.

For 231 the condition is $F_2 = E_2 \oplus E_3$, the regular zero locus of $\mathcal{F}_2 \rightarrow \mathcal{E}_1$. Its class is $c_2(\mathcal{F}_2^\vee \otimes \mathcal{E}_1) = z_1 z_2$. For 312 the condition is $F_1 = E_3$, the regular zero locus of $L_1 \rightarrow \mathcal{E}_1 \oplus \mathcal{E}_2$. Its class is $z_1(z_1 + t)$. These loci are smooth copies of a projective line in each fiber. On the latter locus, imposing $F_2 = E_2 \oplus E_3$ is the regular zero locus of $L_2 \rightarrow \mathcal{E}_1$, with class z_2 . The projection formula gives the class $z_1 z_2(z_1 + t)$ for the opposite Schubert point 321. This also fixes the complex orientations used for these classes.

Definition 7.1. The genuine Schubert lift $\sigma_w \in H_G^{\ell(w)\lambda}(X)$ is the unique class with $q_X(\sigma_w) = u^{\ell(w)} S_w$.

Existence and uniqueness follow from Proposition 5.1. The preceding geometric calculations and (6.1) give Table 2. Thus the basis in the main theorem is the normalized Schubert basis, rather than an arbitrary choice of generators for the additive splitting.

TABLE 2. Normalized Schubert lifts.

w	Notation	Degree	Representative
123	1	0	1
213	A	λ	x_1
132	B	λ	$x_1 + x_2 + a$
231	C	2λ	x_1x_2
312	D	2λ	$x_1(x_1 + a)$
321	F	3λ	$x_1x_2(x_1 + a)$

7.2. Products.

Proposition 7.2. *The multiplication of the nonunit Schubert classes is given in Table 3. Together with R -linearity and the coefficient multiplication, this determines the full ring.*

Proof. Use the two reductions

$$(7.1) \quad x_1^3 = a^2x_1, \quad x_2^2 = -x_1x_2 - x_1^2 + a^2$$

from (6.3). For example, $A^2 = D - aA$, and

$$AB = x_1^2 + x_1x_2 + ax_1 = C + D.$$

Substituting $x_2 = B - A - a$ in the second reduction gives $B^2 = C - aB$. Multiplication of $C = x_1x_2$ by A gives $AC = F - aC$, while $AD = x_1^3 + ax_1^2 = aD$. For the other divisor products, reduction gives

$$BC = x_1^2x_2 + x_1x_2^2 + ax_1x_2 = aC, \quad BD = F - aD.$$

Also $AF = BF = aF$, by the same reductions. The remaining entries follow from these formulas without any division:

$$\begin{aligned} C^2 &= C(B^2 + aB) = a^2C + a^2C = -a^2C, \\ D^2 &= D(A^2 + aA) = -a^2D, \\ CD &= (CA)(A + a) = (F - aC)(A + a) = aF, \\ CF &= C^2(A + a) = -a^2F, \\ DF &= (DC)(A + a) = aF(A + a) = -a^2F, \\ F^2 &= (FC)(A + a) = a^3F. \end{aligned}$$

All signs are in $\mathbb{F}_3$. Each reduction is an equality in the genuine ring presentation, including when coefficients lie in N . $\square$

The three identities (1.4) are the divisor, or Monk–Chevalley-type, formulas in this example. All structure constants have the form ca^r with $c \in \mathbb{F}_3$ and $r \geq 0$. Indeed the coefficient of σ_z in $\sigma_v\sigma_w$

TABLE 3. Schubert multiplication; the lower triangle follows by commutativity.

$\cdot$	A	B	C	D	F
A	$D - aA$	$C + D$	$F - aC$	aD	aF
B		$C - aB$	aC	$F - aD$	aF
C			$-a^2C$	aF	$-a^2F$
D				$-a^2D$	$-a^2F$
F					a^3F

has degree $(\ell(v) + \ell(w) - \ell(z))\lambda$ and is constrained by Lemma 2.1. The negative cone contributes additional classes but no additional Schubert structure constants.

8. FIXED POINTS AND THE FAILURE OF NAIVE GKM

8.1. Restriction matrix and localization. At p_w , the line quotient L_i has character χ^{w_i-1} . Naturality of (6.1), and the coefficient comparison in degree λ , imply

$$(8.1) \quad x_i|_{p_w} = r_{w_i}a, \quad (r_1, r_2, r_3) = (0, -1, 1).$$

Substitution in Table 2 gives the complete restriction matrix in Table 4.

The diagonal entries $1, -a, -a, -a^2, -a^2, a^3$ agree with the oriented Euler products of the representations W_w . For instance $W_{231} = \chi^2 \oplus \chi$ contributes $(-a)a = -a^2$, and $W_{321} = \chi^2 \oplus \chi \oplus \chi^2$ contributes a^3 .

Let T denote the displayed matrix, with basis elements indexing its columns. It factors as

$$(8.2) \quad T = M \operatorname{diag}(1, a, a, a^2, a^2, a^3), \quad M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}.$$

The matrix M has determinant 1 in $\mathbb{F}_3$, whereas $\det T = a^9$.

Corollary 8.1. *Fixed-point restriction induces an isomorphism of graded rings*

$$H_G^*(X)[a^{-1}] \xrightarrow{\cong} \prod_{w \in S_3} R[a^{-1}].$$

Proof. Both factors of (8.2) become invertible after inverting a . Restriction is a ring map, so the resulting module isomorphism is also a ring isomorphism. $\square$

8.2. The kernel and image before localization.

Proposition 8.2. *At G/G , the kernel is*

$$\ker i^* = J_1A \oplus J_1B \oplus J_2C \oplus J_2D \oplus J_3F.$$

It is a nonzero square-zero ideal. In particular,

$$(8.3) \quad 0 \neq \mu_{1,1}A \in H_G^2(X), \quad i^*(\mu_{1,1}A) = 0.$$

TABLE 4. Restriction to the six fixed flags.

p_w	1	A	B	C	D	F
123	1	0	0	0	0	0
213	1	$-a$	0	0	0	0
132	1	0	$-a$	0	0	0
231	1	$-a$	a	$-a^2$	0	0
312	1	a	$-a$	0	$-a^2$	0
321	1	a	a	$-a^2$	$-a^2$	a^3

Proof. Invert M in (8.2), without inverting a . The kernel is exactly that of the diagonal factor, yielding the stated direct sum. By (2.6) all its coefficients belong to N , and $N^2 = 0$, so products of two kernel elements vanish. The class $\mu_{1,1}$ and the relation $a\mu_{1,1} = 0$ are part of (2.2)–(2.4), obtained from [BG21, Proposition 3.4]. Freeness of the Schubert basis makes $\mu_{1,1}A$ nonzero. Its degree is $(2 - \lambda) + \lambda = 2$. $\square$

Proposition 8.3. *A homogeneous tuple $f = (f_e, f_{s_1}, f_{s_2}, f_{231}, f_{312}, f_{321}) \in (R^\alpha)^6$ belongs to $\text{im } i^*$ if and only if*

$$(8.4) \quad \begin{aligned} f_e - f_{s_1} &\in aR, \\ f_e - f_{s_2} &\in aR, \\ f_e + f_{s_1} - f_{s_2} - f_{231} &\in a^2R, \\ f_e - f_{s_1} + f_{s_2} - f_{312} &\in a^2R, \\ f_{321} - f_e + f_{s_1} + f_{s_2} - f_{231} - f_{312} &\in a^3R. \end{aligned}$$

Membership in $a^r R$ here means membership in $a^r R^{\alpha-r\lambda}$.

Proof. Direct inversion of M gives

$$M^{-1}f = \begin{pmatrix} f_e \\ f_e - f_{s_1} \\ f_e - f_{s_2} \\ f_e + f_{s_1} - f_{s_2} - f_{231} \\ f_e - f_{s_1} + f_{s_2} - f_{312} \\ f_{321} - f_e + f_{s_1} + f_{s_2} - f_{231} - f_{312} \end{pmatrix}.$$

The tuple lies in the image of T exactly when these coordinates are divisible by $1, a, a, a^2, a^2, a^3$, respectively. For sufficiency choose a preimage for each coordinate under the corresponding power of a , and use those preimages as Schubert coefficients. Their possible nonuniqueness is precisely the kernel in Proposition 8.2. $\square$

8.3. Root curves. For each transposition (ij) and flag p_w , the subgroup acting unitarily on $E_{w_i} \oplus E_{w_j}$ produces a root curve $\mathbb{CP}^1$ joining p_w to $p_{w(ij)}$. Its tangent character at p_w is $\chi^{w_j - w_i}$, and the character at the other end is its inverse. There are nine such curves. These are the torus-invariant root curves, not an enumeration of all curves invariant under the finite group.

Each root curve is equivariantly a representation sphere S^λ , up to the orientation choice. Its two fixed-point restrictions have the form $(c, c \pm ad)$. This follows from $H_G^*((S^\lambda)_+) \cong R \oplus R[-\lambda]$: the reduced generator restricts to zero at the basepoint and to the Euler class at the other fixed point. Thus the condition from every root curve is

$$(8.5) \quad f_v - f_w \in aR.$$

The ideals generated by a and $-a$ agree.

Corollary 8.4. *The ordinary edge-divisibility conditions do not determine the image of the genuine fixed-point restriction map for X .*

Proof. In our fixed-point order take

$$f = (0, 0, 0, 0, 0, a) \in (R^\lambda)^6.$$

Every difference of two entries is zero or $\pm a$, so (8.5) holds for all root curves. The last expression in (8.4), however, equals a and does not belong to a^3R . Indeed a homogeneous preimage would have degree -2λ , and $R^{-2\lambda} = 0$. $\square$

In pure multiples of λ , restriction itself is injective: Schubert coefficients lie in $\mathbb{F}_3[a]$, on which the diagonal factor of (8.2) is injective. Thus Corollary 8.4 is not merely a consequence of the negative-cone kernel. The higher divisibilities already constrain the image in these pure degrees. The classical rational torus GKM presentation [Kaj15, §2.3] uses the ideals of individual roots. In the present calculation the root Euler factors are all $\pm a$.

9. MACKEY STRUCTURE AND CONSISTENCY CHECKS

9.1. Coefficient restriction and transfer. At the trivial orbit the coefficient ring is

$$\underline{R}^*(G/e) = \mathbb{F}_3[v^{\pm 1}], \quad |v| = \lambda - 2.$$

In a particular degree α this notation records the ordinary coefficient group in dimension $|\alpha|$; distinct powers of v are distinct $RO(G)$ degrees of underlying dimension zero. Restriction is

$$(9.1) \quad \text{res}(u) = v, \quad \text{res}(a) = \text{res}(\kappa) = 0, \quad \text{res}(N) = 0.$$

For a and κ this follows from the underlying coefficient degrees 2 and 1. For $\nu_{j,k}$ the underlying degree is $1 - 2k$, and for $\mu_{j,k}$ it is $2 - 2k$; only $\mu_{j,1}$ could have nonzero restriction. Its restriction is zero, as the following sphere calculation also shows.

Lemma 9.1. *The negative-cone generators can be normalized compatibly with (2.4) so that*

$$(9.2) \quad \text{tr}(v^n) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1}, & n < 0. \end{cases}$$

Proof. For $n \geq 0$, Frobenius reciprocity and $\text{tr}(1) = 3 = 0$ give $\text{tr}(v^n) = u^n \text{tr}(1) = 0$. For $j \geq 1$, use the cofiber sequence

$$S(j\lambda)_+ \longrightarrow S^0 \longrightarrow S^{j\lambda}.$$

Its cohomology connecting map is an isomorphism in degree $2j - 1$ to reduced degree $2j$ on $S^{j\lambda}$, both at G/G and at G/e , since the adjacent positive integer coefficients of a point vanish. The sphere $S(j\lambda)$ is free, with quotient the oriented lens space $L^{2j-1}(3)$. Integer-graded cohomology with constant coefficients agrees here with ordinary cohomology of that quotient. On this free space, restriction and transfer are the pullback and cohomological transfer of the three-sheeted covering

$$S^{2j-1} \longrightarrow L^{2j-1}(3).$$

On top integral cohomology pullback is multiplication by 3, whereas transfer sends the orientation generator to the orientation generator: evaluation of a transferred cocycle on the base fundamental class equals evaluation on its lifted fundamental class. Reducing modulo 3, pullback is zero and transfer is nonzero. Naturality of the connecting maps gives the same assertion on $\widetilde{H}_G^{2j}(S^{j\lambda})$.

Desuspending by $j\lambda$ gives target degree $2j - j\lambda$, with $R^{2j-j\lambda} = \mathbb{F}_3\mu_{j,1}$, and underlying generator v^{-j} . Thus (9.1) holds and

$$\text{tr}(v^{-j}) = c_j \mu_{j,1}, \quad c_j \in \mathbb{F}_3^\times.$$

For $j > 1$, Frobenius reciprocity gives $u \text{tr}(v^{-j}) = \text{tr}(v^{1-j})$. Since $u\mu_{j,1} = \mu_{j-1,1}$, we obtain $c_j = c_{j-1}$. Writing c for this common scalar, replace every $\nu_{j,k}$ and $\mu_{j,k}$ by $c\nu_{j,k}$ and $c\mu_{j,k}$, respectively. This preserves all the relations in (2.4) and gives (9.2). $\square$

The vanishing of $\text{tr}(1)$ does not force the transfers in all representation degrees to vanish.

9.2. The flag cohomology as a graded Green functor. Let $y_i = c_1(L_i^\vee) \in H^2(X; \mathbb{F}_3)$. The value at G/e is

$$(9.3) \quad \underline{H}_G^*(X)(G/e) = \frac{\mathbb{F}_3[v^{\pm 1}, y_1, y_2, y_3]}{(e_1(y), e_2(y), e_3(y))}, \quad |y_i| = 2, \quad |v| = \lambda - 2.$$

Restriction on coefficients is (9.1), and on the ring generators it is $\text{res}(x_i) = vy_i$. If $S_w(y)$ denotes the ordinary Schubert polynomial, define

$$\bar{\sigma}_w = \text{res}(\sigma_w) = v^{\ell(w)} S_w(y).$$

This bar notation includes the displayed power of v . Then

$$(9.4) \quad \text{tr}(v^n \bar{\sigma}_w) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1} \sigma_w, & n < 0. \end{cases}$$

Indeed Frobenius reciprocity gives $\text{tr}(b \text{res } \sigma_w) = \text{tr}(b) \sigma_w$, and Lemma 9.1 applies. Since the $\bar{\sigma}_w$ form a basis over $\mathbb{F}_3[v^{\pm 1}]$, this specifies every transfer.

The Weyl action on (9.3) is trivial: the action on X comes from a connected torus and is homotopic to the identity, and G preserves all representation orientations. The Mackey double-coset formula consequently reads

$$\text{res tr} = 1 + g + g^2 = 0,$$

as is also immediate from (9.1) and (9.4). Frobenius reciprocity for arbitrary products follows from the coefficient formulas and the Schubert basis. To see the compatibility explicitly, all terms in the Schubert multiplication involving a positive power of a vanish both upon restriction and upon multiplication by $\mu_{j,1}$. The remaining terms are the ordinary Schubert products. Coefficient reciprocity also holds on the generators: $u\mu_{j,1} = \mu_{j-1,1}$, $a\mu_{j,1} = \kappa\mu_{j,1} = 0$, and $N\mu_{j,1} = 0$. These observations check the formulas on an additive basis and hence on all elements. Together with Theorem 1.1, they specify the graded Green functor.

The G/e value of fixed-point restriction sends every y_i to zero and sends v to the same v in all six factors. Its image is the diagonal copy of $\mathbb{F}_3[v^{\pm 1}]$ and its kernel is (y_1, y_2, y_3) . This lower-orbit kernel is not square zero; for example $y_1^2 \neq 0$. Thus the square-zero assertion in Proposition 8.2 is specifically a statement at G/G .

9.3. Ordinary specialization and the top class. Forgetting the action and the representation bookkeeping sends a, κ, N to zero and sends u, v to the ordinary unit. It gives

$$H^*(X; \mathbb{F}_3) \cong \mathbb{F}_3[y_1, y_2, y_3]/(e_1(y), e_2(y), e_3(y)).$$

The ordinary degrees of the basis are 0, 2, 2, 4, 4, 6, of total rank 6 = $|S_3|$. The class F restricts, with its grading factor, to $v^3 y_1^2 y_2$. On the bundle $\text{Fl}(\mathbb{C}^3) \rightarrow \mathbb{CP}^2$ selecting the first line, y_2 integrates to 1 on the $\mathbb{CP}^1$ fiber and y_1 is the positive hyperplane class on the base. Therefore

$$\int_X y_1^2 y_2 = 1.$$

So F specializes to the top Schubert class with the complex orientation. Finally, Table 3 gives

$$A^2 B = F + a(D - C), \quad AB^2 = F + a(C - D).$$

After setting $v = 1$, the underlying classes satisfy $\bar{A}^2 \bar{B} = \bar{A} \bar{B}^2 = \bar{F}$. The genuine formula $F^2 = a^3 F$ is also consistent with its restriction a^3 at 321 and zero at the other five flags.

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